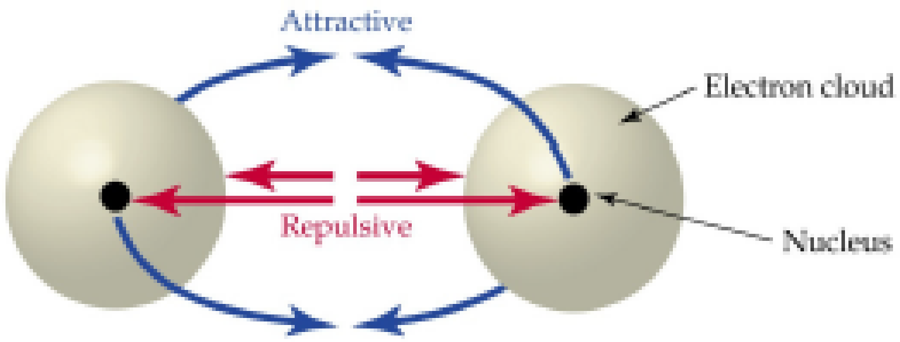
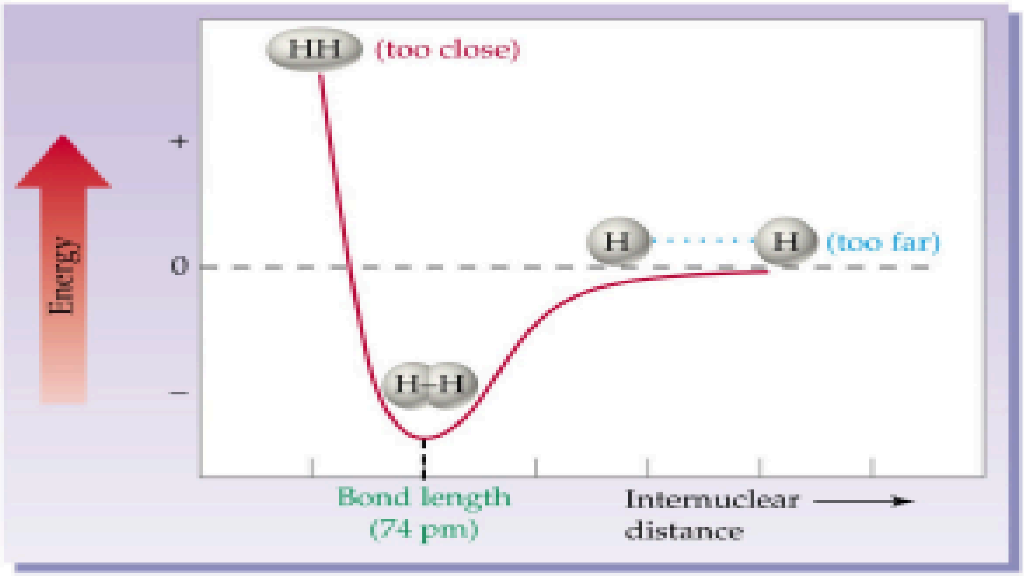
Lec(7):Valence Bond Theory (VBT) and Hybridization

A covalent H-H bond is the net result of attractive and repulsive electrostatic forces. When bringing together two atoms that are initially very far apart. Three types of interaction occur:



(1) The nucleus-electron attractions (blue arrows) are greater than the (2) nucleus-nucleus and (3) electron-electron repulsions (red arrows), resulting in a net attractive force that holds the atoms together to form an H_2 molecule.



A graph of potential energy versus internuclear distance for the H_2 molecule.

•If the hydrogen atoms are too far apart, attractions are weak and no bonding occurs. A zero of energy when two H atoms are separated by great distances. A drop in potential energy (net attraction) as the two atoms approach each other.

•When the atoms are optimally separated, the energy is at a minimum. A minimum in potential energy at particular internuclear distance (74pm) corresponding to the stable H_2 molecule and the potential energy corresponds to the negative of the bond dissociation energy.

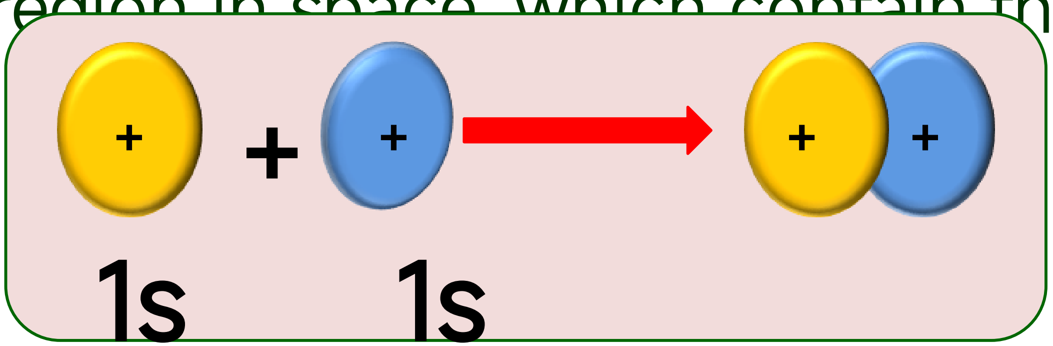
If the atoms are too close, strong repulsions occur. A increase in potential energy as the atoms approach more closely.

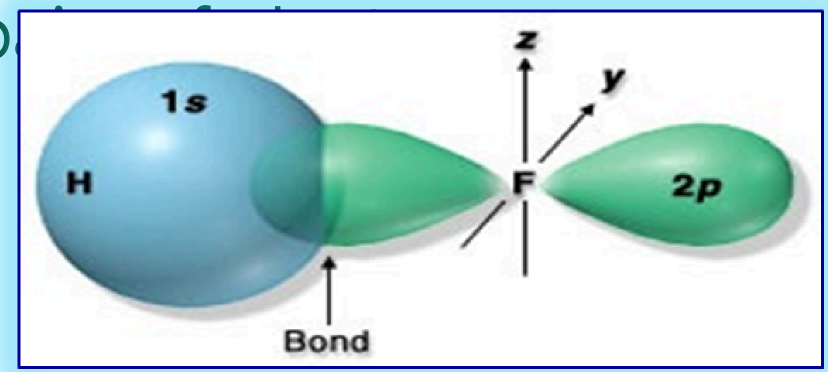
Valence Bond Theory (VBT)

* **Bond formation by overlapping orbitals**: A description of covalent bond formation in terms of atomic orbital overlap is called **valence bond theory**. It gives a localized electron model of bonding: **core electrons** and **lone-pair valence electrons** retain the same orbital locations as in the separated atoms, and the charge density of the bonding electrons is concentrated in the region of orbital overlap. **VBT** focuses on how the atomic orbitals of the dissociated atoms combine on molecular formation to give individual chemical bonds. ***The basic***

The main postulates of Valence Bond Theory (VBT)

1- When two atoms come together to form a covalent bond, an atomic orbital of one atom *overlaps* with an atomic orbital of the other. By overlap, the two orbitals share some common region in space, which contain the p





2- The strength of the covalent bond is proportional to ***the degree of overlap.***

" The greater the overlap, the stronger the bond ."

Hybridization of Atomic Orbitals

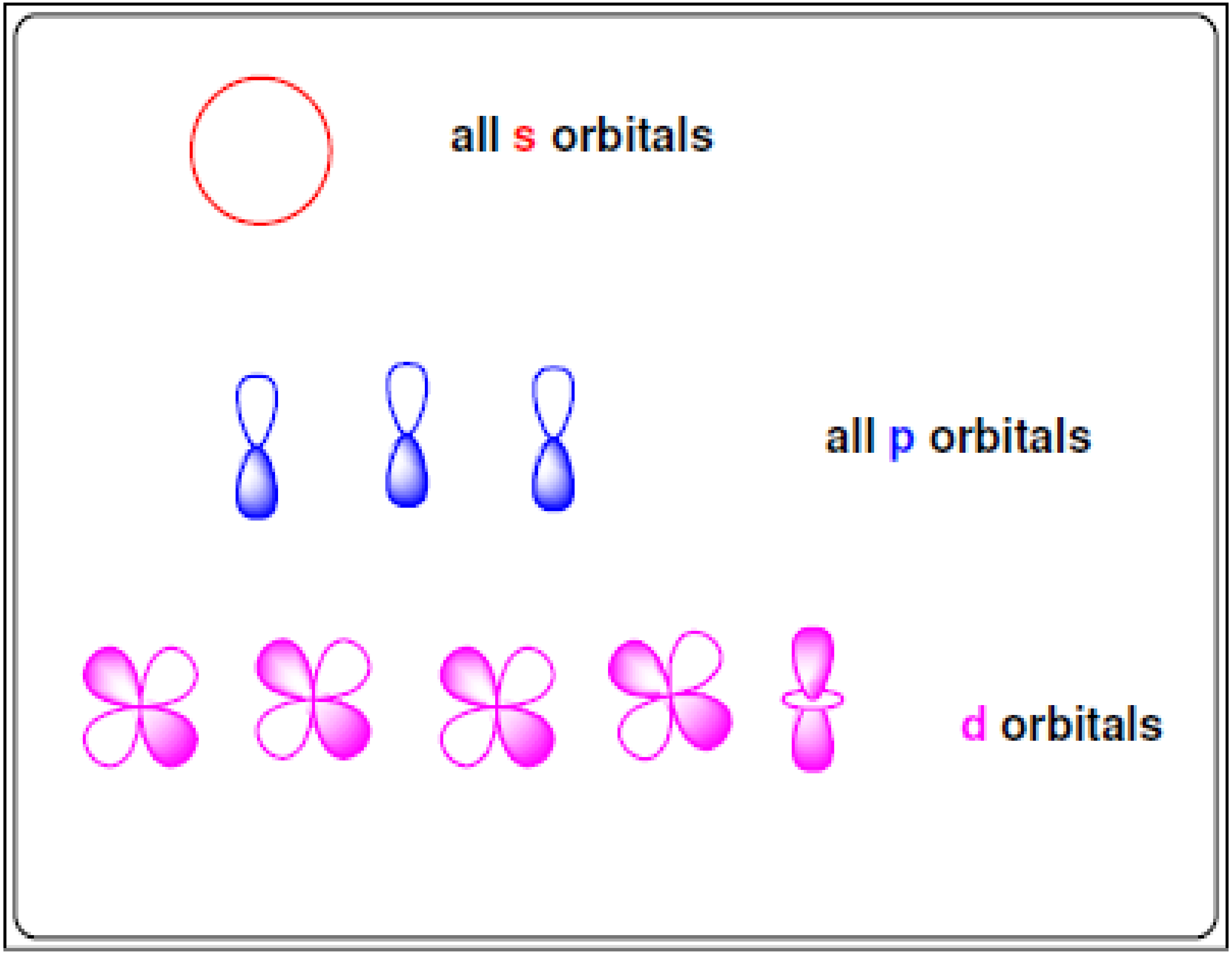
Linus Pauling (in 1931), proposed that the wave function for the s and p atomic orbitals can be mathematically combined to form a new set of equivalent wave function called **hybrid orbitals**.

The mathematical process of replacing pure atomic orbitals with reformulated atomic orbitals for bonded atoms is called **hybridization**. It means mixing of the atomic orbitals of an isolated atom (central atom) to generate a new set of equivalent atomic orbitals called "**hybrid orbitals**". In hybridization scheme, the number of **hybrid orbitals** equals to the total number of

Basic concepts of Hybridization

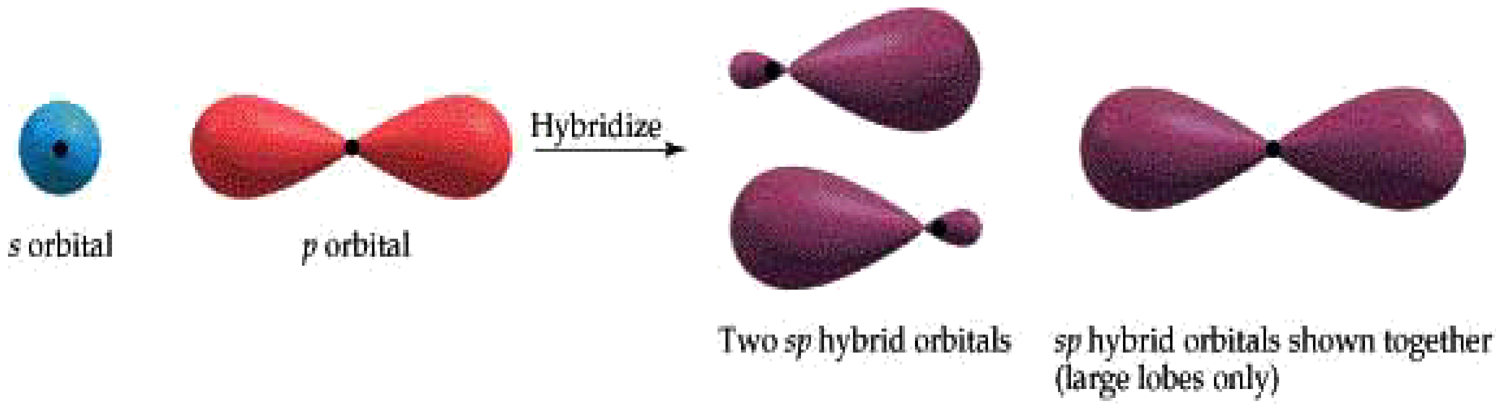
- The number of hybrid orbitals is equal to the number of mixed atomic orbitals.
- All orbitals in a set of hybrid orbitals are equivalent "have the same energy and the same shape".
- Hybrid orbitals have shapes and orientations that are completly different from those of the atomic orbitals in isolated atom.
- Hybrid orbitals do not exist in isolated atoms. They are found only in covalently bonded atoms.
- The shape of the molecule is determined by sigma bond and lone pairs.
- Sigma bonds in polyatomic molecules usually involve the overlap of hybrid orbitals. While, pi bond result from the overlap of unhybridized orbitals.

Orbital shapes, Individual ("isolated") Atoms



Formation of sp hybrid orbitals

The combination of an s orbital and a p orbital produces 2 new orbitals called sp orbitals.



These new orbitals are called hybrid orbitals

The process is called hybridization

What this means is that both the s and one p orbital are involved in bonding to the connecting atoms

sp *Hybridization* Linear Arrangement

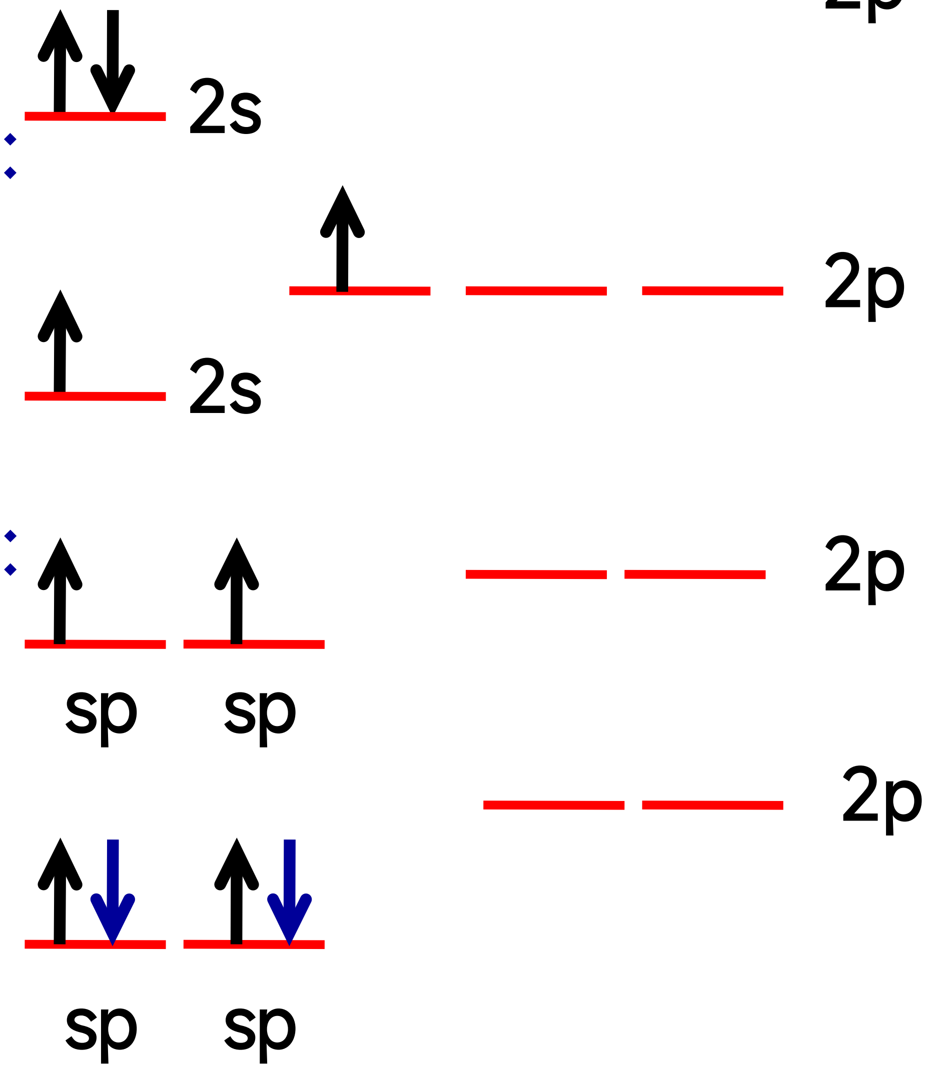
Example of sp *Hybridization*: BeH_2

^{4}Be: $1s^2$, $2s^2$

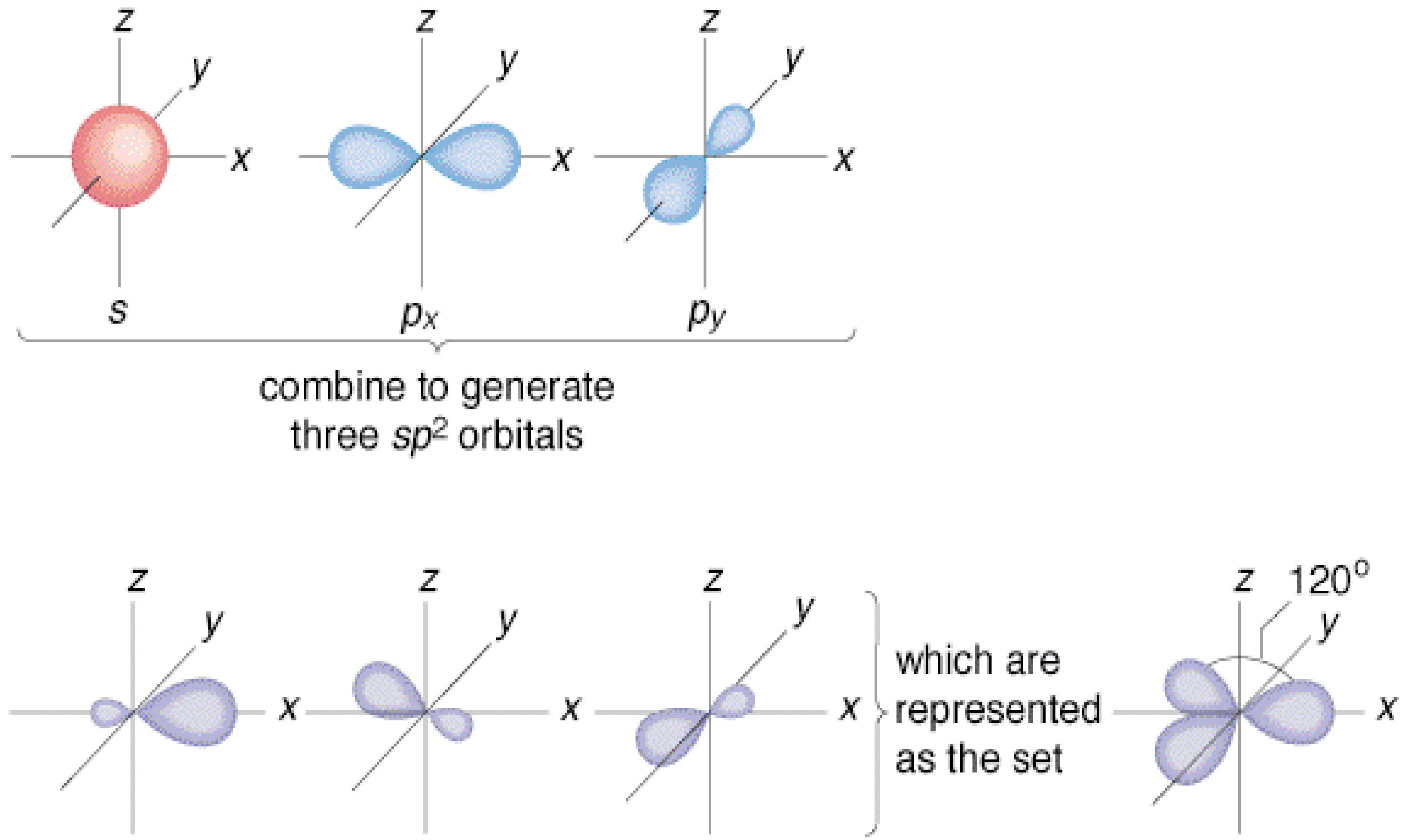
^{4}Be atom in ground state:

^{4}Be atom in excited state:

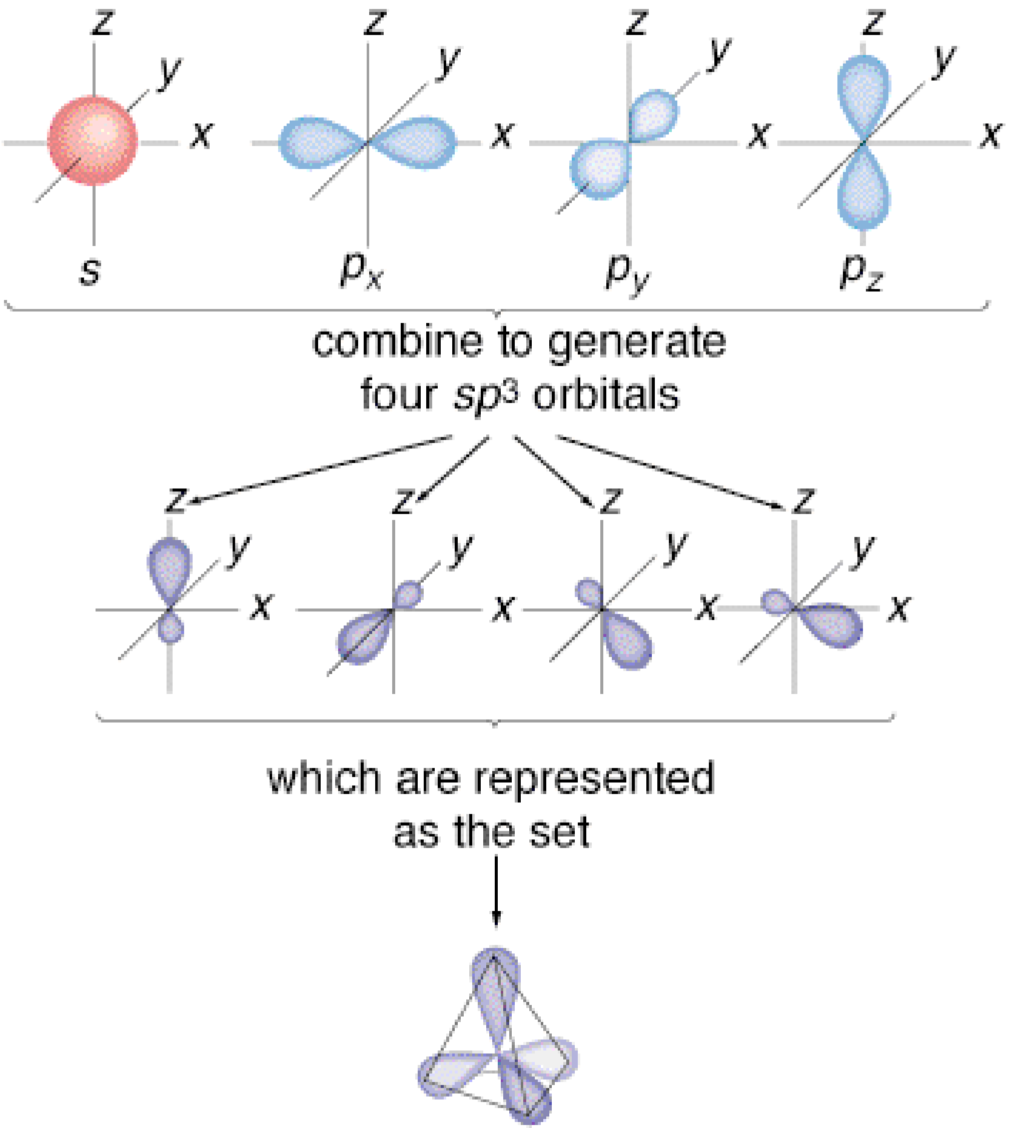
sp Hybridization:



Formation of sp^2 hybrid orbitals



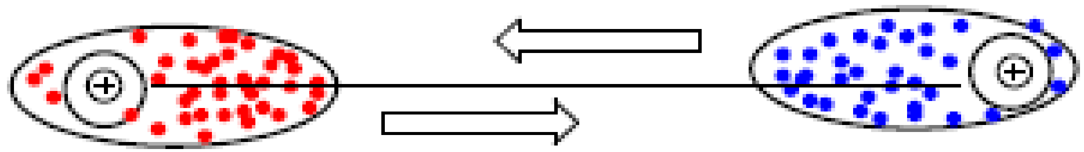
Formation of sp³ hybrid orbitals



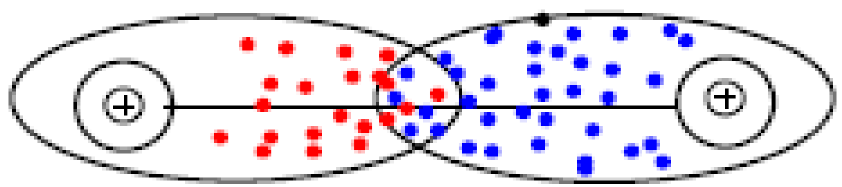
COVALENT BOND FORMATION (VB THEORY)

In order for a covalent bond to form between two atoms, overlap must occur between the orbitals containing the valence electrons.

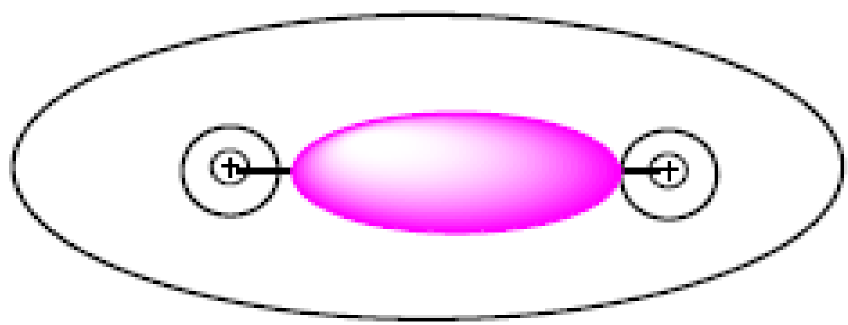
The best overlap occurs when two orbitals are allowed to meet "head on" in a straight line. When this occurs, the atomic orbitals merge to form a single bonding orbital and a "single bond" is formed, called a sigma (σ) bond.



Dotted areas: representation of "electron cloud" for one electron



"Head-on Overlap"



Sigma Bond: merged orbital, 2 e's

MAXIMIZING BOND FORMATION

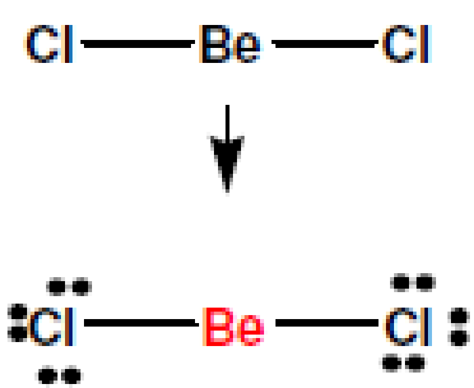
In order for "best overlap" to occur, valence electrons need to be re-oriented and electron clouds reshaped to allow optimum contact.

To form as many bonds as possible from the available valence electrons, sometimes separation of electron pairs must also occur.

We describe the transformation process as "orbital hybridization" and we focus on the central atom in the species...

“sp” Hybridization: all 2 Region Species

$BeCl_2$

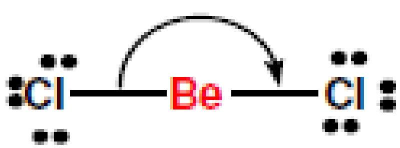


Be	2
2Cl	14

16 e's/2= 8 prs

(octet violator)

Number of regions around CENTRAL ATOM: 2



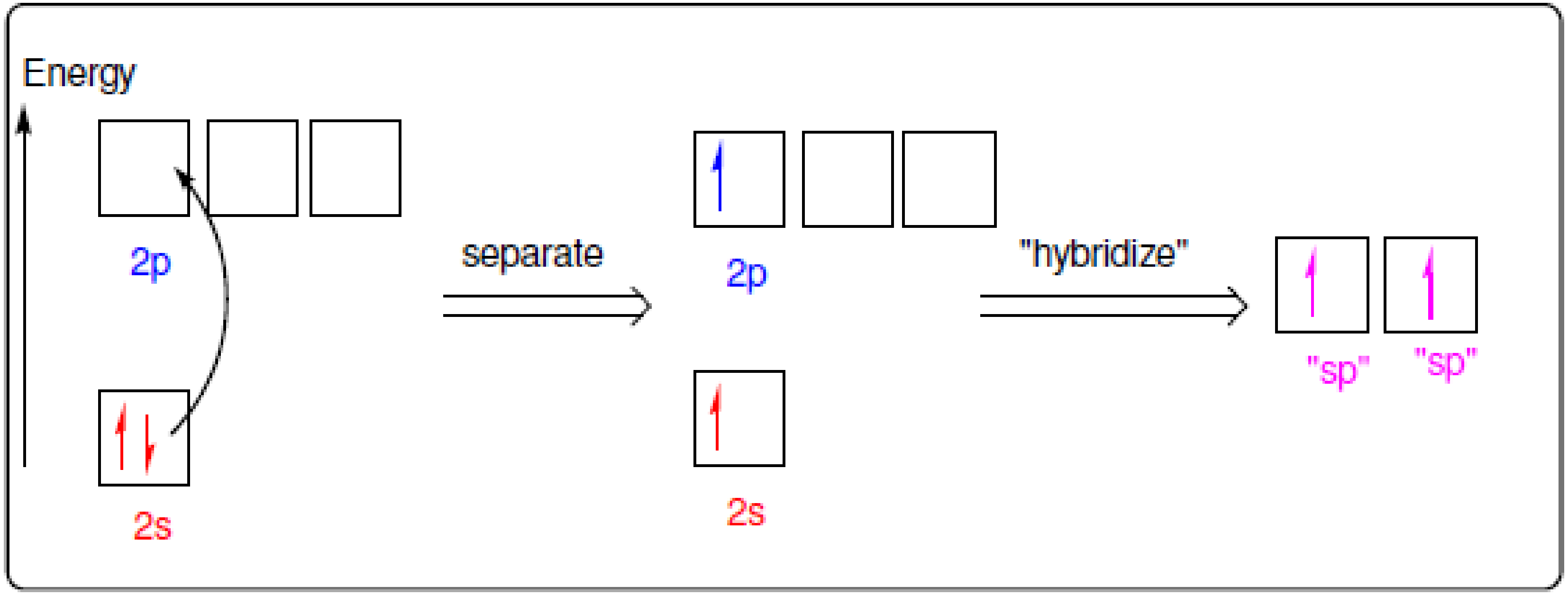
shape : LINEAR
bond angles: 180°

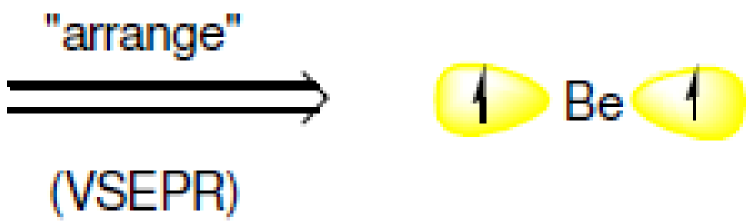
Hybridization of Be in $BeCl_2$

Valence e's

Atomic Be: $1s^2$ $2s^2$

Hybrid sp orbitals:
1 part s, 1 part p

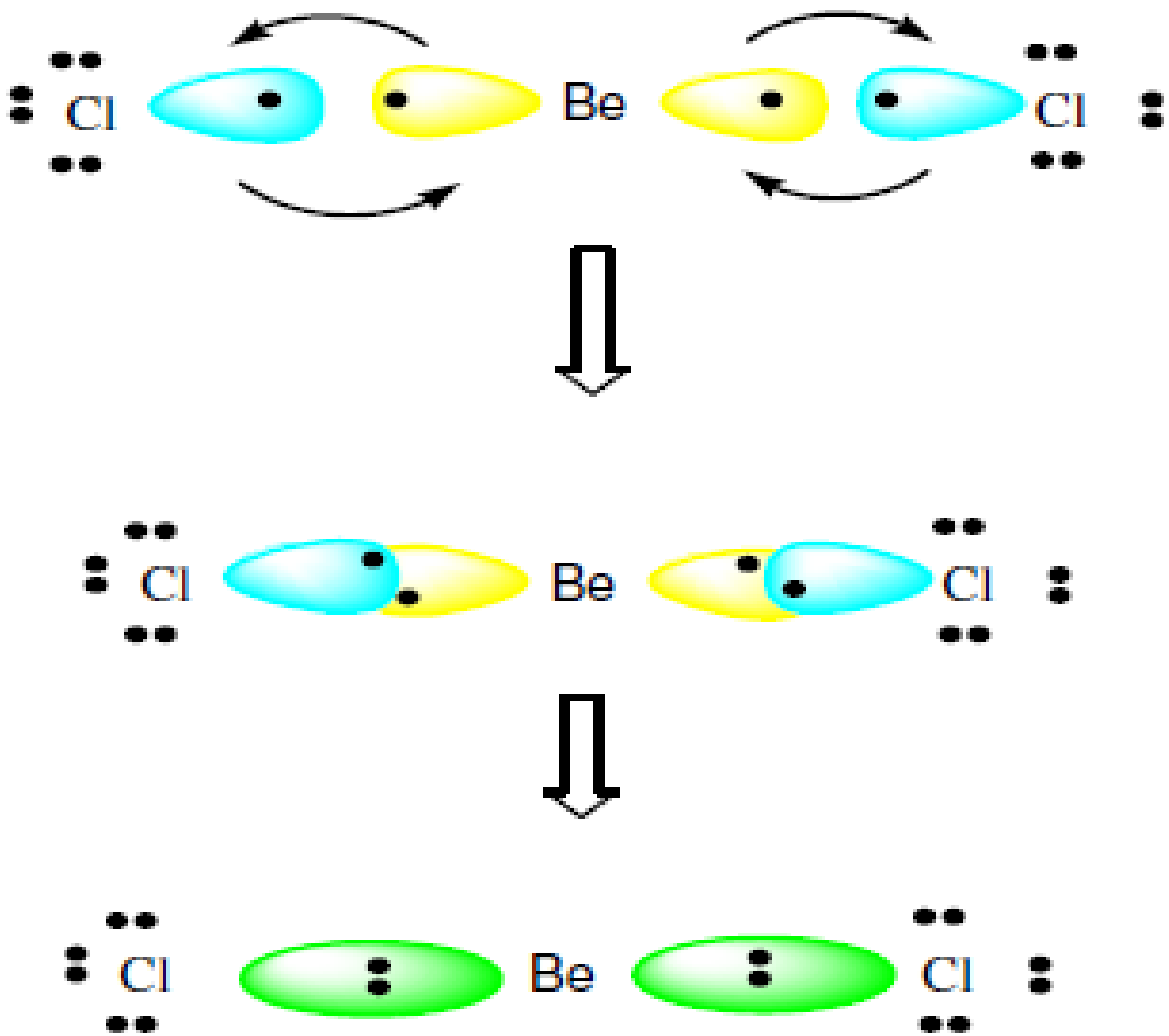




Be is said to be "sp hybridized"

FORMATION OF $BeCl_2$:

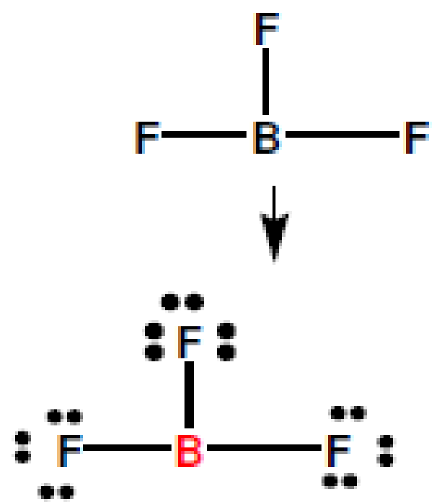
Each Chlorine atom, $1s^2 2s^2 2p^6 3s^2 3p^5$, has one unshared electron in a p orbital. The half filled p orbital overlaps head-on with a half full hybrid sp orbital of the beryllium to form a sigma bond.



sp hybridized, linear, 180° bond angles

"sp^2" Hybridization: All 3 Region Species

BF_3

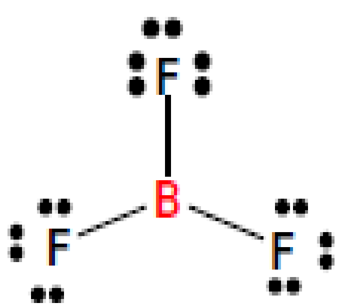


B	3
3F	21

24 e's/2= 12 prs

(octet violator)

Number of regions around CENTRAL ATOM: 3



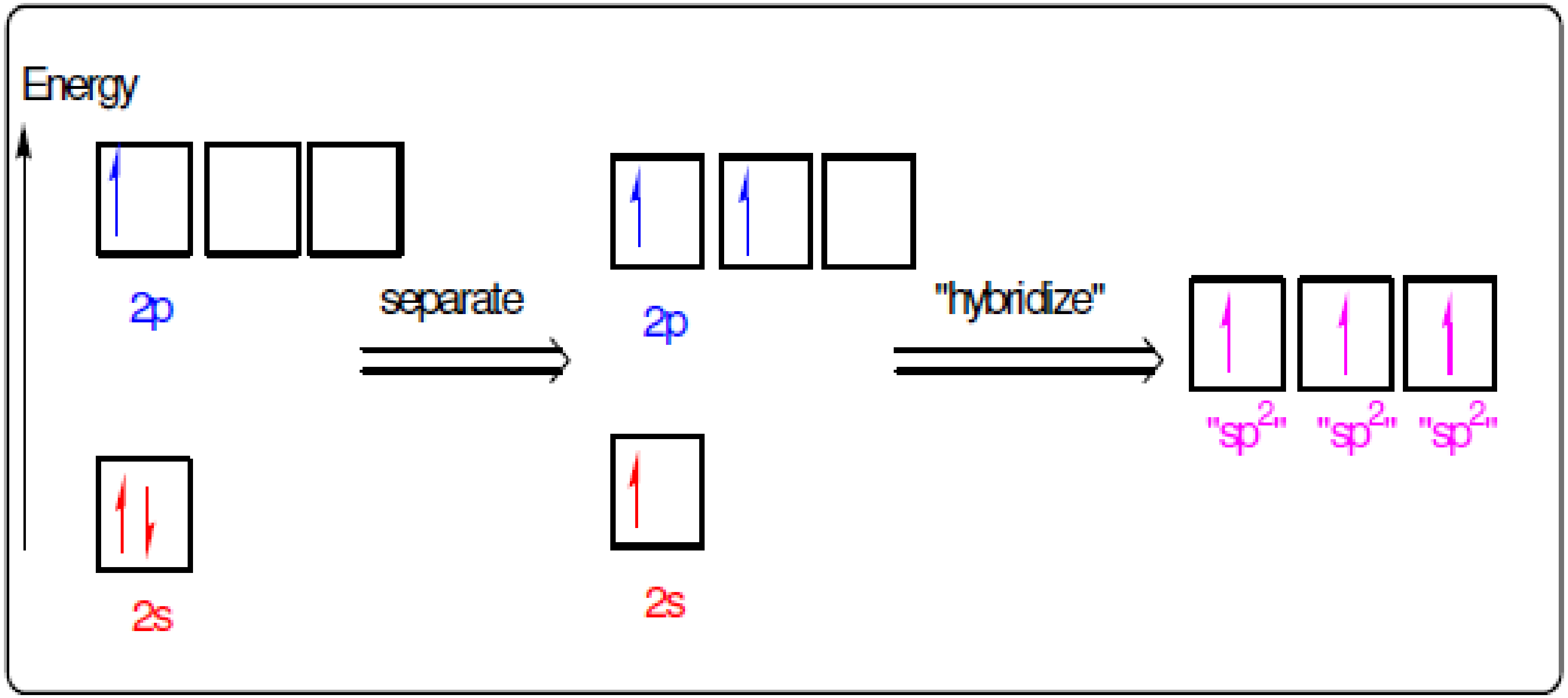
shape : TRIGONAL PLANAR
bond angles: 120°

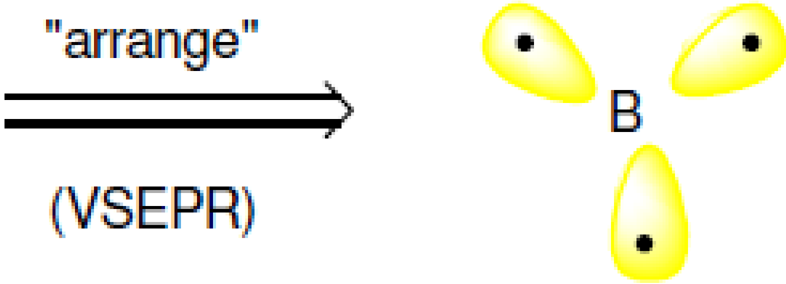
Hybridization of B in BF_3

Valence e's

Hybrid sp^2 orbitals:
1 part s, 2 parts p

Atomic B : $1s^2\ 2s^2\ 2p^1$



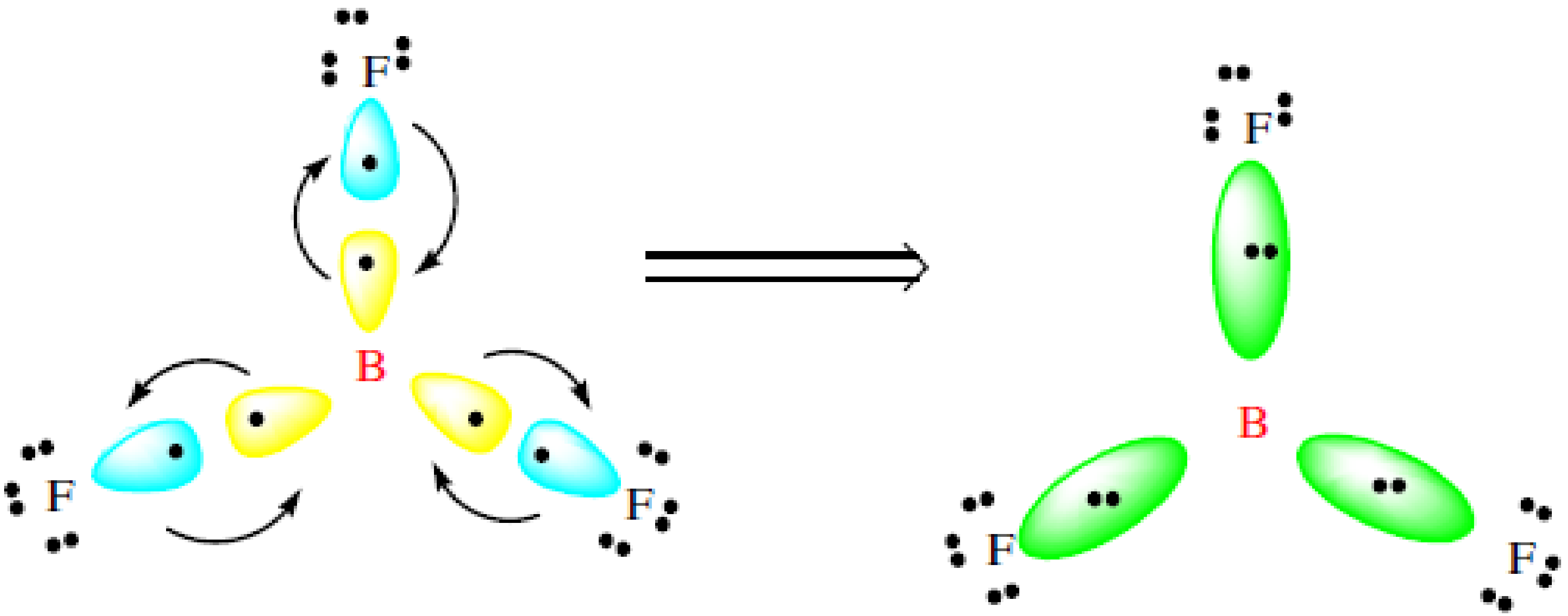


B is said to be "sp^2 hybridized"

FORMATION OF BF_3:

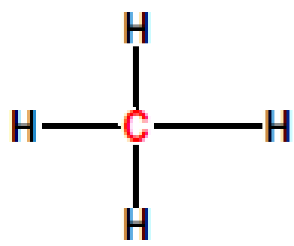
Each fluorine atom, $1s^2 2s^2 2p^5$, has one unshared electron in a p orbital. The half filled p orbital overlaps head-on with a half full hybrid sp^2 orbital of the boron to form a sigma bond.

sp^2 hybridized, TRIGONAL PLANAR,
120^o bond angles



"sp^3" Hybridization: All 4 Region Species

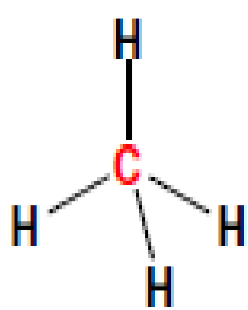
CH_4



C	4
4H	4
	8

8 e's /2 = 4 pr

Number of regions around CENTRAL ATOM: 4



shape : TETRAHEDRAL

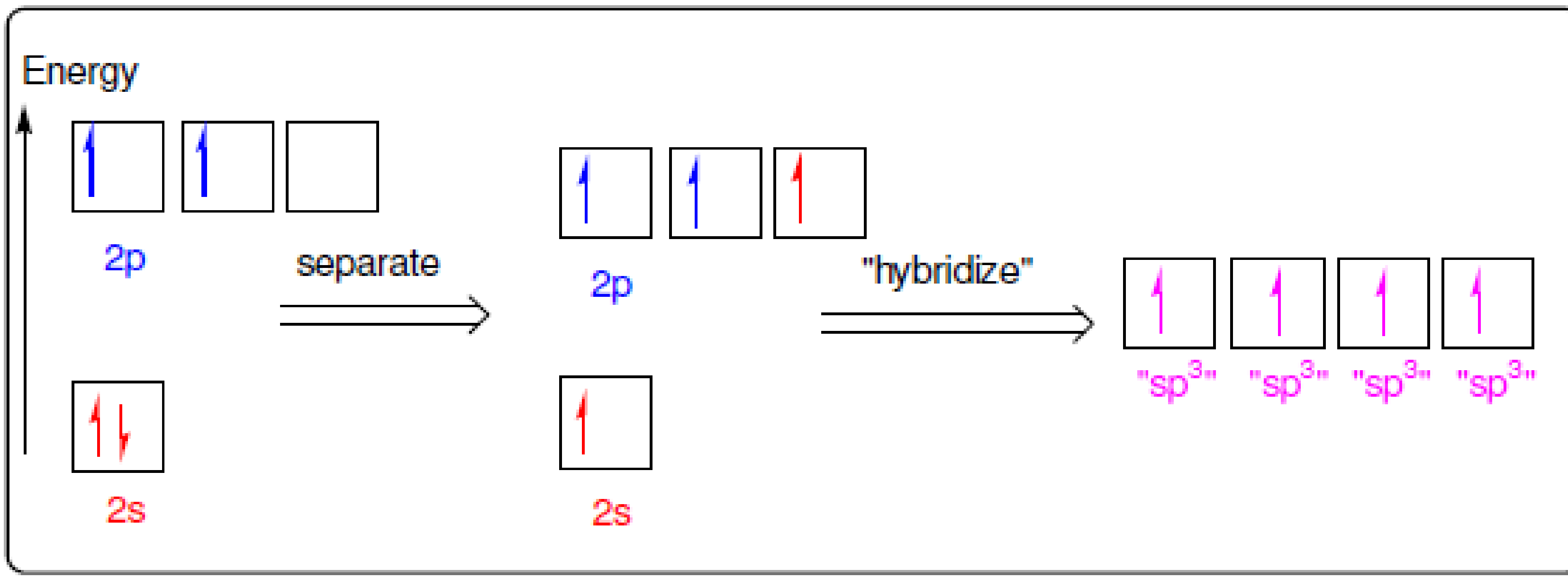
bond angles: 109.5°

Hybridization of C in CH_4

Valence e's

Hybrid sp^3 orbitals: 1 part s, 3 parts p

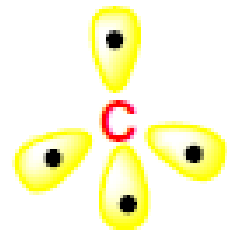
Atomic C : $1s^2\ 2s^2\ 2p^2$



"arrange"

$\Longrightarrow$

(VSEPR)

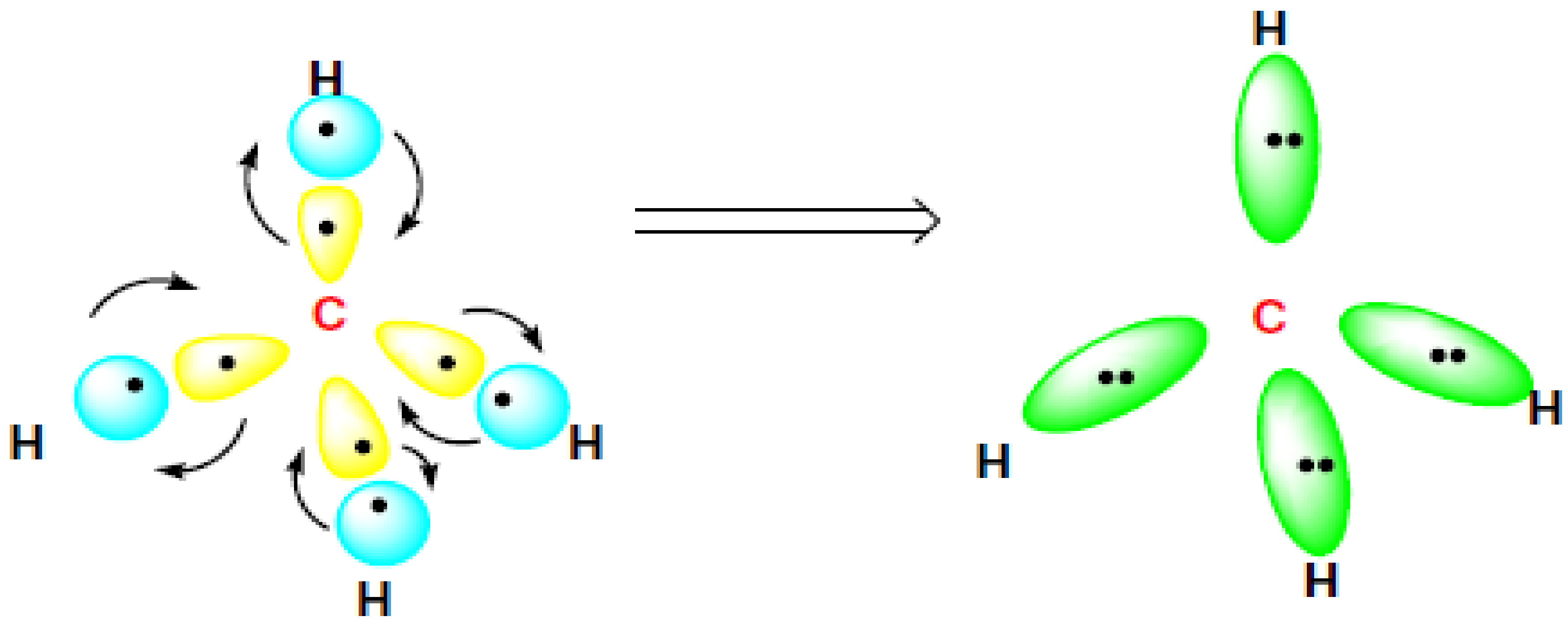


C is said to be
"sp^3 hybridized"

FORMATION OF CH_4:

Each hydrogen atom, $1s^1$, has one unshared electron in an s orbital. The half filled s orbital overlaps head-on with a half full hybrid sp^3 orbital of the carbon to form a sigma bond.

sp^{3}hybridized, TETRAHEDRAL,
109.5° bond angles



sp^3 *Hybridization with lone pair of electrons* (NH_3)

^7N: $1s^2$, $2s^2$, $2p^3$

^7N atom in ground state:

2p: ↑ ↑ ↑

2s: ↑↓

sp^3 Hybridization:

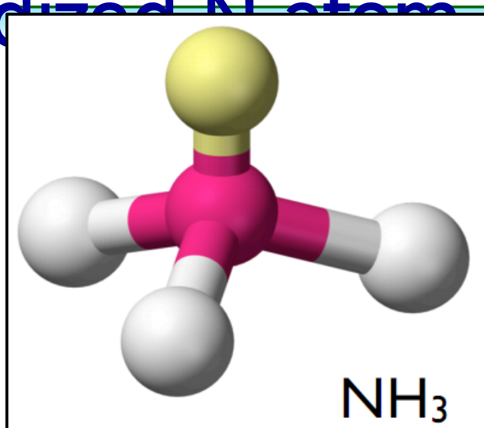
↑↓ ↑ ↑ ↑

sp^3 sp^3 sp^3 sp^3

Hybridized N atom

↑↓ ↑↓ ↑↓ ↑↓

sp^3 sp^3 sp^3 sp^3



sp^3 *Hybridization*

Tetrahedral Arrangement

sp^3 *Hybridization with twp lone pair of electrons* (H_2O)

^8O: $1s^2$, $2s^2$,$2p^4$

^8O atom in ground state:

2s

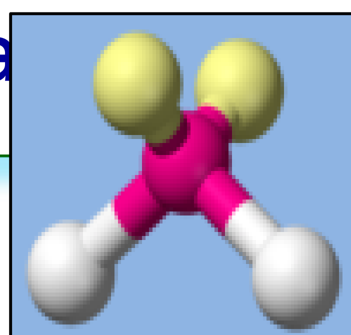
2p

sp^3 Hybridization:

sp^3 sp^3 sp^3 sp^3

Hybridized O atom in H_2O

sp^3 sp^3 sp^3 sp^3



sp^3 *Hybridization*

Tetrahedral Arrangement

sp^3d *Hybridization* PCl_5

Trigonal bipyramid Arrangement

^{15}P: [Ne], $3s^2$,$3p^3$

^{15}P in G. S.

3s 3p 3d

^{15}P in E. S.

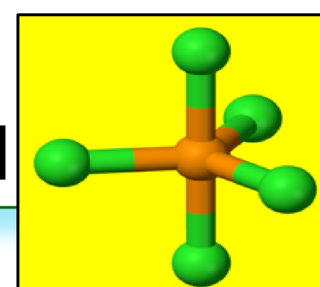
3s 3p 3d

sp^3d sp^3d sp^3d sp^3d sp^3d

sp^3d-Hybridization:

3d

sp^3d sp^3d sp^3d sp^3d sp^3d

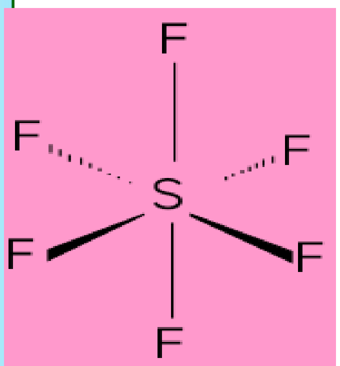


Hybridized P atom

sp^3d^2 Hybridization (SF_6)

Octahedron

Arrangement



^{16}S in G. S.

3s

3p

3d

^{16}S in E. S

3s

3p

3d

Six sp^3d^2 hybridized orbitals

3d

sp^3d^2–Hybridization:

3d

Six sp^3d^2 hybridized orbitals

Hybridized S atom